

Understanding Amateur Actors' VR Rehearsal Needs: Priorities and Rationales

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Abstract

This study explores which needs actors prioritize in a VR rehearsal environment for amateur theatre and the reasons behind these priorities. In particular, it examines which types of VR rehearsal supports amateur actors consider most important and how they explain these choices. A mixed methods survey was conducted with amateur actors from a university theatre community. Participants completed a forced ranking task across six VR rehearsal support themes and explained their preferences through open-ended questions. The forced ranking data were combined using Borda Count to obtain a group level priority order, while open-ended responses were evaluated using codebook thematic analysis. This study provides insights into how VR rehearsal supports are prioritized in an amateur theatre context.

CCS Concepts

• **Human-centered computing** → **Virtual reality**; *Interaction design*; Collaborative interaction; • **Arts and humanities** → Performance studies.

Keywords

Virtual Reality, Co-presence, Theatre Rehearsal, Amateur Theatre, Embodied Interaction

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1 INTRODUCTION AND BACKGROUND

Theatre rehearsal depends on actors' ability to explore space, timing, and interaction while staying responsive to scene partners and maintaining performance meaning in the moment [13]. When rehearsals must be conducted remotely or under limited access to stage resources, virtual reality (VR) can provide a shared rehearsal medium in which performers can explore embodied aspects of staging, including blocking and spatial relationships [6, 8, 10, 22].

However, effective VR rehearsal is not achieved through visual simulation alone. Rehearsal usefulness depends on whether performers feel situated in the virtual space and whether interaction is credible and responsive [16, 17]. Theatre training also highlights sustained attention and connection with scene partners (e.g., Stanislavski's notion of "communion") [20]. Accordingly, VR rehearsal design should prioritize actor-centered supports for partner-oriented action, communication, and expressive social cues (e.g., gesture, gaze, turn-taking) over visual realism alone [3–5, 18, 23].

At the same time, current platforms constrain expressive embodiment (e.g., missing facial expressions and limited gesture detail) [6, 15, 22]. Since rehearsal involves repeated exposure, comfort issues such as cybersickness and eye strain become important for design [19]. Designers must therefore decide which rehearsal supports to prioritize while keeping systems accessible and usable [10, 14].

Most VR rehearsal research has focused on professional theatre contexts, while amateur theatre remains widely practiced yet underrepresented in academic research [11, 12, 21]. Amateur rehearsal settings may involve distinct constraints, such as limited access to rehearsal space, restricted schedules, and varying technical familiarity, which can shift which VR supports are relevant and practical.

To address this gap, this study characterizes how amateur actors prioritize VR rehearsal needs and why, based on a forced-ranking task and thematic analysis of open-ended responses. Specifically, we address the following research questions: *RQ1: How are six VR rehearsal support themes prioritized?* and *RQ2: What rationales and expectations are expressed for these priorities?*

Participants completed a forced-ranking task across six VR rehearsal support themes and answered open-ended questions. Rankings were aggregated using Borda Count [1], and responses were analyzed with codebook thematic analysis [2].

2 METHODS

2.1 Recruitment of Participants and Data Collection

Forty amateur actors (20 women, 20 men) were recruited from Özyeğin University through a university-affiliated theatre community. Participation was voluntary, and informed consent was obtained through the online survey. The participants were university students from various disciplines and did not receive formal theatre education. These participants were members of an amateur theatre group within the university and were involved in community-level theatre practice rather than professional training. Before joining stage productions, they completed basic acting training. Trainee



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Theme	Function	Key features
Comfort & Spatial Usability	Configuring a comfortable and usable VR rehearsal space.	Adjustable lighting; sufficient movement space; visual comfort.
Social Interaction & Co-presence	Supporting responsive rehearsal interaction with others in VR.	Multi actor co-presence; NPC partners; asynchronous audio recordings.
Script & Directorial Guidance	Supporting rehearsal through script interaction and director led guidance.	In script note-taking; highlighting; director annotations; personalized layout.
Stage Layout, Set & Props	Supporting blocking and staging through virtual set and prop representations.	Placement of decorations; prop representations; stage layout simulation.
Practice & Feedback Tools	Supporting individual practice, memorization, and self improvement.	Repetition support; memory games; drawing/writing tools; self review recordings.
Rehearsal Logistics & Continuity	Reducing disruptions and supporting continuity across rehearsal sessions.	Supporting rehearsals despite space constraints; facilitating coordination; maintaining rehearsal flow.

Figure 1: VR rehearsal support themes and brief descriptions provided to participants.

members who had not completed this training were excluded. Only participants with experience in at least one amateur stage production were included in the study.

The six themes were based on a prior study with amateur actors [7]. Their labels and descriptions were revised using relevant literature [6, 8, 10, 22], while the overall structure remained the same. The themes were refined to make them clearer, reduce overlap, and ensure they were suitable for amateur actors, including those without prior VR experience. The final set of themes was used for the ranking task. Each theme was presented with a short description explaining its purpose and key features (see Fig. 1), so that participants understood them in the same way before completing the ranking. Participants completed a web-based survey that included demographic questions, a forced-ranking task of the six VR rehearsal support themes (1 = most important, 6 = least important; no duplicate ranks), and open-ended questions about their choices.

2.2 Data Analysis

This study analyzed the forced-ranking data using the Borda Count method [1] to aggregate individual rankings into a single group-level priority order. With six themes, rank positions were assigned fixed weights ($1^{st}=6$, $2^{nd}=5$, $3^{rd}=4$, $4^{th}=3$, $5^{th}=2$, $6^{th}=1$). For each theme, a total Borda score was computed by summing these weights across all participants ($N = 40$), and themes were ordered from highest to lowest score. Higher scores indicate that a theme was ranked closer to the top more consistently across participants, reflecting aggregated priorities within the sample.

Open-ended responses were analyzed using Codebook Thematic Analysis [2] to interpret participants' rationales for the quantitative rankings. A deductive approach was used, with the six predefined rehearsal support themes serving as the initial coding framework. Responses were coded at the semantic level, focusing on the explicit meaning of participants' statements. All responses were coded by a single researcher following a predefined codebook. The codebook was developed following guidelines [9], including brief and full definitions as well as inclusion and exclusion criteria to reduce overlap across themes. Responses containing multiple distinct ideas were assigned to more than one theme when appropriate, and representative quotes were selected to illustrate recurring rationales and contextualize the Borda results.

3 PRELIMINARY RESULTS

This study recruited 40 participants (see Fig. 2) with balanced gender representation and varying levels of VR experience (55% experienced; 45% no experience). Acting experience ranged from 0–3 years to over 5 years, with the largest group reporting 5 or more years of experience. Participants were predominantly young adults (17–27 years), with the largest subgroup aged 23–25.

3.1 Borda Count Aggregation

The Borda count aggregation provided an overall prioritization of the six rehearsal support themes (see Fig 3). Social Interaction & Co-presence received the highest total Borda score (170 points), indicating that participants placed the strongest emphasis on immersive and interactive aspects of the VR rehearsal experience. This was followed by Stage Layout, Set & Props (149 points), suggesting that staging-related support and the integration of physical elements into the virtual rehearsal workflow were also perceived as highly valuable.

A second tier of themes emerged in the mid-range of the ranking. Script & Directorial Guidance (136 points) and Rehearsal Logistics & Continuity (135 points) were nearly tied, indicating that participants considered script-related support and rehearsal problem management to be comparably important. Finally, Comfort & Spatial Usability (127 points) and Individual Practice & Feedback Tools (123 points) received lower total scores relative to the other themes, suggesting that while these features were still valued, they were less frequently prioritized at the top of participants' rankings.

Overall, participants prioritized interaction and staging features over secondary support features. However, scores for the mid- and lower-ranked themes were relatively close, so the ordering should be interpreted as a general tendency rather than a strict separation.

3.2 Codebook Thematic Analysis

3.2.1 Social Interaction & Co-Presence. Participants tended to value an interactive rehearsal environment as a key condition for useful theatre work. They described interactivity as important for ensemble communication, shared scene-building, and rehearsal realism, suggesting that rehearsal quality depends on co-presence and responsiveness rather than isolated individual practice. As P39 stated: "In a team without communication, the chances of producing a strong and successful play are low." This becomes especially clear in multi-actor scenes, where performers must respond to partners'

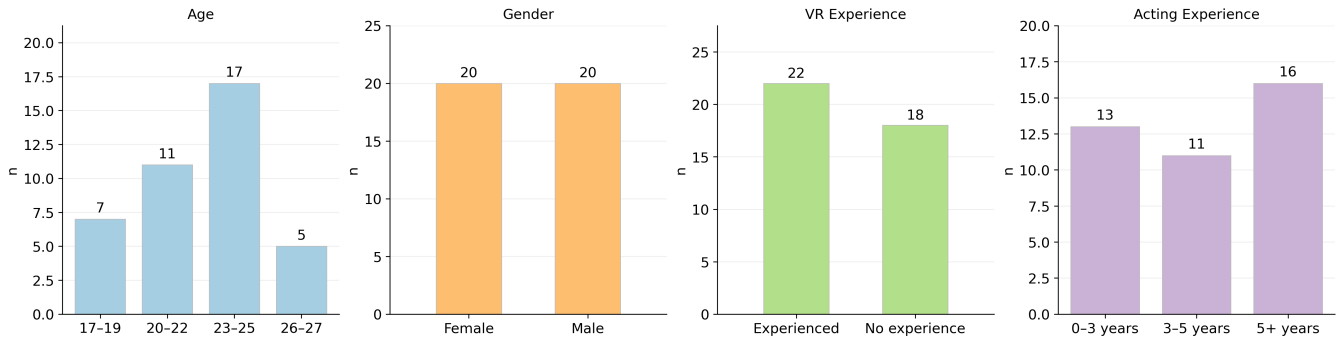


Figure 2: Participant demographics (N=40).

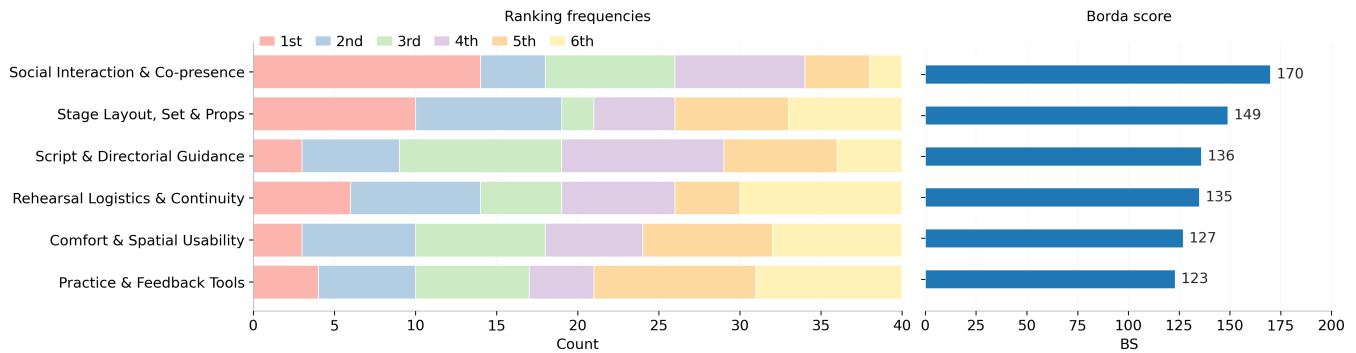


Figure 3: Ranking frequencies (1st–6th) and Borda scores for rehearsal theme priorities (N=40).

lines and movements in real time. P20 explained that Social Interaction & Co-presence can “make the rehearsal process as realistic as possible,” because actors rehearse scenes “while interacting with each other.” In addition, NPC-based rehearsal partners were seen as a way to extend interaction while maintaining immersion. As P26 noted, socially convincing NPCs can help actors behave “as if there is a real person in front of us.” Overall, these points suggest that interactivity supports not only communication and realism during rehearsal, but also more transferable performance preparation for live performance.

3.2.2 Stage Layout, Set & Props. The second-ranked theme, Stage Layout, Set & Props, was valued because it allows stage elements to be used in rehearsals early and many times. This supports directing and blocking by helping actors understand spatial relationships and find their position in the stage environment. Participants said this is important because conventional rehearsals often add set and atmosphere elements late, which gives actors little time to adapt. As P22 noted: “Since the set of a play usually takes its final form only in the later stages of rehearsals, it gives the actor very little time to adapt.” Participants also explained that set design affects performance quality and character work. For example, P19 stated: “the set is just as important. It has a major impact on getting into character.” Therefore, virtual access to the set and props was seen not only as a visual aid, but also as a tool for immersion and early creative decisions.

3.2.3 Script & Directorial Guidance. Script & Directorial Guidance was described as important for structuring the rehearsal process and supporting character creation. Participants explained that script-based work goes beyond memorization and includes dramaturgy and meaning-making within the production. In this context, the script was seen as a central organizing tool, as P3 stated: “I see script and dramaturgy as important elements in managing the play process and character creation.” This theme was also connected to directing through director’s notes and rehearsal guidance. For example, P6 emphasized that “detailed work on the script with director’s notes is very important.” Participants also linked script support to continuity and rehearsal flow, especially when mistakes happen. As P4 noted, if a performer forgets lines, continuing to read can prevent disruption and “keep the play smooth.” Overall, Script & Directorial Guidance was framed as rehearsal support that brings together dramaturgical understanding, director-led instruction, and performance consistency.

3.2.4 Rehearsal Logistics & Continuity. Rehearsal Logistics & Continuity focused on practical constraints that interrupt rehearsal continuity and reduce efficiency in conventional theatre processes, especially issues related to scheduling, space, and coordination. These limits can make it difficult to rehearse consistently as an ensemble. For example, P11 mentioned the “difficulty in gathering everyone at the same time for rehearsals” and “sometimes lack of space for rehearsals,” while P5 pointed to organizational problems such as “gathering the team, stage availability, and reservations.”

These barriers were described as serious obstacles to production, and P9 stated: “Rehearsal difficulties are among the biggest obstacles to staging a play.” Time and cost were also recurring pressures. P12 argued that VR can support rehearsals that are “difficult, time-consuming, and costly... more quickly and with fewer difficulties,” and P16 noted that regular in-person meetings can become unworkable, leading to cancellations when “rehearsals cannot be held and many projects get canceled from the start.” Overall, VR rehearsal was seen as a possible way to reduce ongoing problems in coordination, space, time, and continuity.

3.2.5 Comfort & Spatial Usability. Comfort & Spatial Usability focused on setting up the VR rehearsal space as a basic requirement for a smooth rehearsal process. Participants suggested that if the environment is not comfortable or usable, other parts of rehearsal may also be affected. As P28 stated: “I think that if the environment setting is not suitable, the other aspects will also be disrupted.” In a similar way, P29 linked comfort to efficiency, noting: “If the environment feels comfortable, it can make rehearsals more efficient.” Overall, participants described Comfort & Spatial Usability as supporting rehearsal flow by providing a stable, usable, and comfortable rehearsal environment.

3.2.6 Practice & Feedback Tools. Practice & Feedback Tools were described as practical supports for individual practice, memorization, and performance improvement. Participants highlighted memorization focused features such as gamification (P4) and noted that rehearsal tools should address actors’ memorization needs (P9). In this context, P14 explained that these tools can make memorization “much faster and more memorable.” Another key benefit was the ability to rehearse repeatedly without depending on others. As P18 stated, “being able to rehearse the scene over and over without tiring anyone else besides the actor could be very beneficial,” and P8 similarly noted: “I can improve my performance individually.” Participants also connected these tools to reflective self-development. For example, P38 emphasized the value of customizable memorization support and suggested that being able to watch oneself during rehearsals could lead to better performance, which was previously only possible “with the video.”

4 DISCUSSION

The observed priority pattern indicates that amateur actors frame VR rehearsal less as a primarily visual simulation and more as a shared, interaction-driven activity in which timing, attention, and partner responsiveness determine rehearsal value. This framing is reflected in both the ranking outcomes and participants’ rationales, which emphasized communication, realism through interaction, and real-time adjustment with scene partners. From an acting perspective, rehearsal depends on sustaining meaning and responsiveness in the moment rather than repeating text in isolation [13].

Accordingly, the top ranking of *Social Interaction & Co-presence* suggests a preference for conditions that preserve ensemble work and mutual adaptation, which are difficult to reproduce through individual practice alone. This aligns with presence research arguing that immersive environments are useful when users feel located in the virtual place and experience interaction as credible and responsive [16, 17]. This also resonates with acting traditions that

emphasize commitment to the given circumstances as a basis for believable action [20]. Together, these perspectives suggest that rehearsal value emerges from interaction plausibility and partner-oriented action rather than representational realism alone. Social VR research similarly emphasizes the role of avatar, interaction, and environment design [4]. Prior VR rehearsal studies further show that embodied co-presence can support collaboration and affective connection [6, 22].

The high placement of *Stage Layout, Set & Props* indicates that spatial understanding is treated as an early rehearsal resource rather than a late-stage add-on. Participants’ rationales emphasized the challenge of adapting to set and stage constraints when these become available only in later rehearsals, suggesting that early spatial access can reduce uncertainty in blocking and positioning. This is consistent with VR rehearsal research showing that actors and directors use virtual environments to explore blocking and spatial relationships before physical stage conditions are available [6, 10]. More broadly, these findings suggest that rehearsal quality depends on acting within a stable spatial frame, which allows actors to test timing, distance, and movement choices that later become part of performance memory [13].

Mid-ranked priorities, *Script & Directorial Guidance* and *Rehearsal Logistics & Continuity*, point to a complementary form of rehearsal value focused on sustaining flow across sessions under amateur constraints. While co-presence and staging support the enactment of scenes in VR, guidance and continuity features appear to stabilize rehearsal workflows by helping actors maintain progress despite scheduling pressure, limited rehearsal space, and coordination difficulties commonly reported in amateur contexts [12, 21]. Prior VR rehearsal work also suggests that tools and rehearsal practices co-evolve through iterative use, implying that guidance and continuity may be most valuable as scaffolding for repeated rehearsal rather than as standalone functions [8, 22].

Finally, the lower ranking of *Comfort & Spatial Usability* and *Practice & Feedback Tools* should not be interpreted as low relevance. Instead, participants’ rationales suggest that these themes function primarily as enabling conditions that reduce friction once the core social and spatial experience is in place, and that become most visible when absent or under repeated exposure. This interpretation is compatible with known constraints in expressive embodiment and usability in current VR platforms, where design effort often involves balancing practical accessibility with technical limits [14, 15]. In this regard, the cost of VR headsets should also be considered a practical barrier, as many amateur and community theatre groups operate with limited budgets and may not be able to afford such equipment. Unlike professional companies, amateur groups often rely on personal funds or small grants, which makes the purchase of multiple headsets for ensemble rehearsal a major concern for real-world adoption. Moreover, comfort-related constraints such as cybersickness and visual fatigue can become particularly salient under repeated rehearsal exposure [19]. Taken together, these results suggest that actor-centered VR rehearsal value is most likely to emerge when co-presence and spatial staging are supported first, while guidance, continuity, and comfort features stabilize the rehearsal workflow over time.

5 CONCLUSION AND FUTURE WORK

This study examined how amateur actors prioritize VR rehearsal support needs and why, using a forced ranking task aggregated with Borda Count and a codebook thematic analysis of open ended responses. Overall, participants prioritized supports that enable rehearsal as a shared and embodied process, particularly *Social Interaction & Co-presence* and *Stage Layout, Set & Props*. Mid ranked priorities emphasized sustaining rehearsal flow through *Script & Directorial Guidance* and *Rehearsal Logistics & Continuity*, while *Comfort & Spatial Usability* and *Practice & Feedback Tools* were described as enabling conditions for effective rehearsal. Limitations include recruitment from a single amateur theatre community and the forced ranking design, which may compress differences between closely valued themes. Variation in participants' VR familiarity also suggests that some priorities may reflect anticipated usefulness rather than sustained rehearsal experience. Finally, all data were coded by one researcher, which may introduce interpretive bias. Future work will validate these priorities through hands on VR rehearsal sessions and examine how prioritization shifts across rehearsal phases and production contexts.

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